

✓ Time-series Econometrics: Vector Auto-Regression (VAR) Models

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Note: This lecture note covers Ch 13 (Linear Time Series Models) and Ch 14 (Structural Vector Autoregressions) of Econometric Modelling with Time Series: Specification, Estimation and Testing by V. L. Martin, A. S. Hurn and D. Harris.

Motivation

A Simple Example

Model Specification

Estimation

Forecasting with Reduced-form VAR

Impulse-Response analysis with Structural VAR

Granger Causality

Implementation in Python

Outline

Motivation

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Estimation

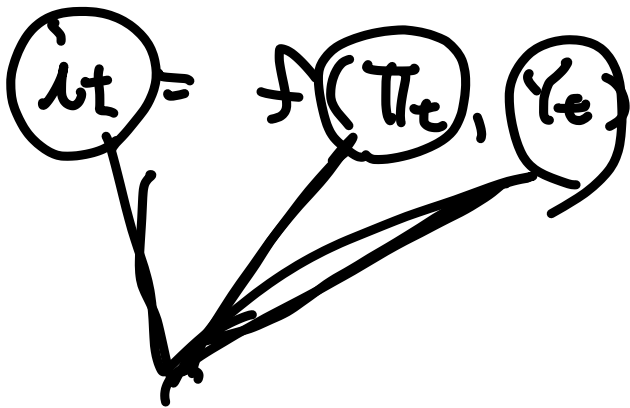
Forecasting with Reduced-form VAR

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Christopher A. Sims



A hand-drawn diagram representing a structural equation model. It consists of three circles. The first circle on the left contains the expression λ_t . The second circle in the middle contains the expression $f(\pi_t, \gamma_t)$. The third circle on the right contains the expression y_t . Arrows point from the first circle to the second, and from the second circle to the third, indicating a causal or functional relationship between the variables.

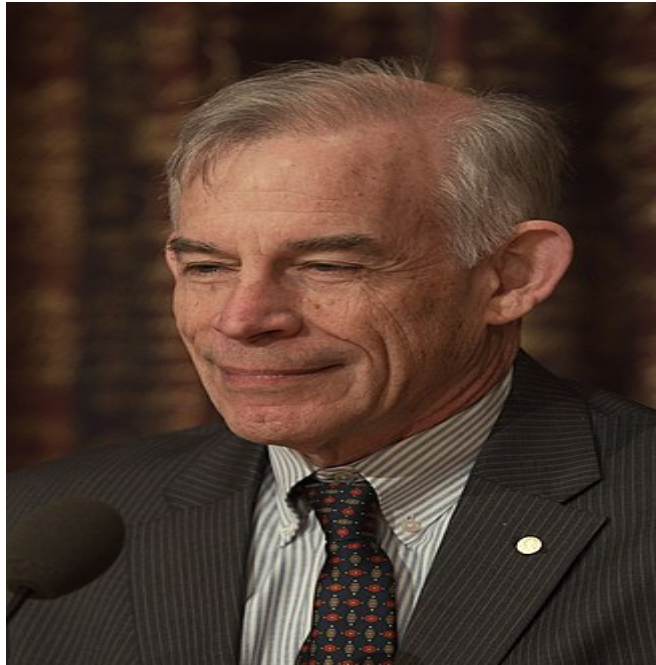


Figure: Christopher A. Sims

- ▶ Christopher Albert Sims (born October 21, 1942) is an American econometrician and macroeconomist. He is currently a Professor of Economics at Princeton University.
- ▶ Together with Thomas Sargent, he won the Nobel Memorial Prize in Economic Sciences in 2011.

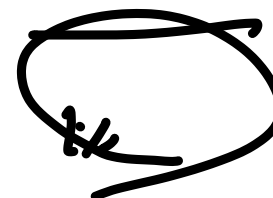
Motivation

Consider n stationary time series variables $\{y_{1t}\}, \dots, \{y_{nt}\}$. A multivariate time series is the $(n \times 1)$ vector time series $\{\mathbf{Y}_t\}$ where the i^{th} row of $\{\mathbf{Y}_t\}$ is $\{y_{it}\}$. That is, for any time t , $\mathbf{Y}_t = (y_{1t}, \dots, y_{nt})'$.

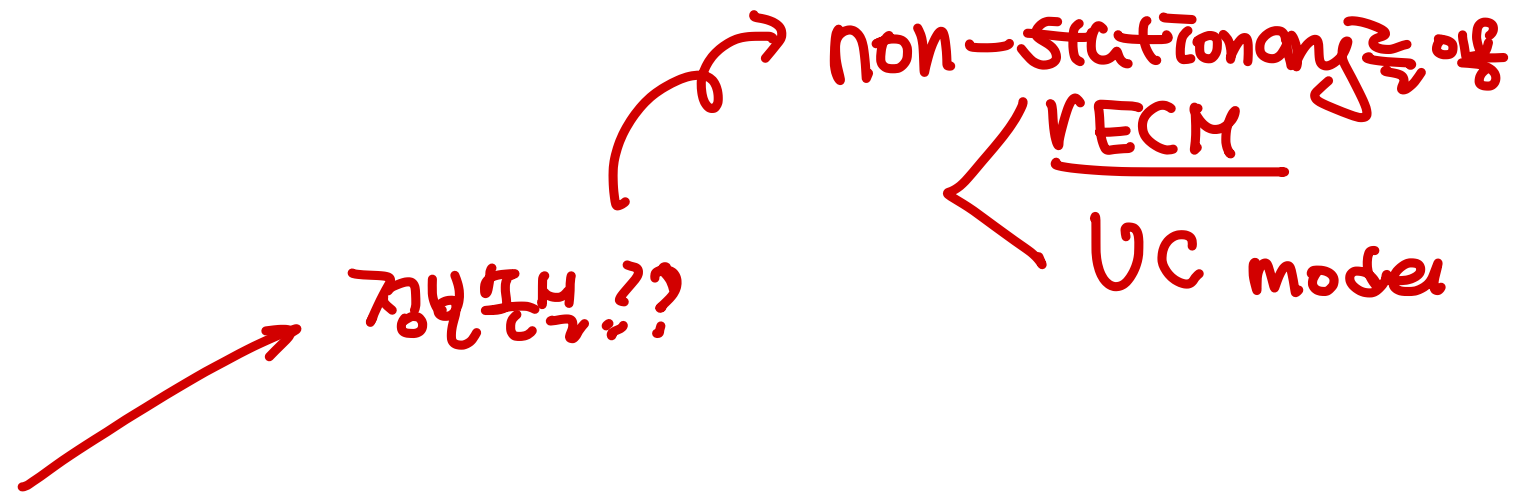
$$\mathbb{E}(e_t | \mathbf{x}_s) = 0$$

Multivariate time series analysis is used when one wants to model and explain the interactions and co-movements among a group of time series variables:

- ▶ Consumption and income
- ▶ Stock prices and dividends
- ▶ forward and spot exchange rates
- ▶ interest rates, money growth, income, inflation
- ▶ etc



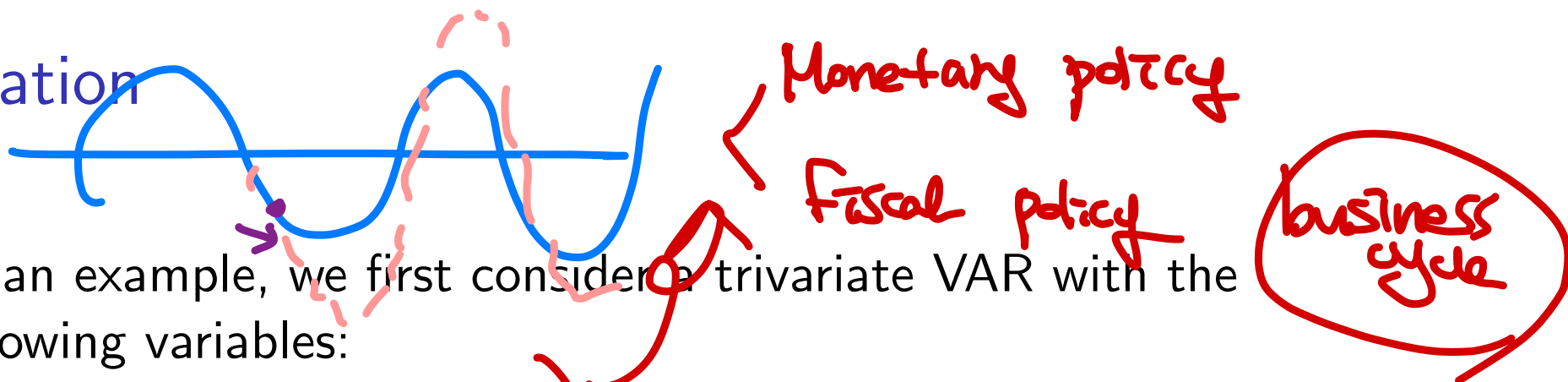
Motivation



Note: VAR models are for stationary time series variables.

Thus, we need to conduct unit root tests before using a VAR model to check if all variable we would like to include are stationary.

Motivation



As an example, we first consider a trivariate VAR with the following variables:

- * Unemployment rate (u_t)
- * Inflation rate (π_t)
- * Policy rate (r_t)

$$\underline{\lambda_t} = \gamma_t + \pi_t^e$$

A VAR can help us answering the following questions

- 1 What is the dynamic behavior of these variables? How do these variables interact? (statistical characteristics)
- 2 What is the most likely path of unemployment rate in the next few quarters? (prediction)
- 3 What is the effect of a monetary policy shock on unemployment rate? (causal inference)

Motivation

What do we expect from a VAR model?

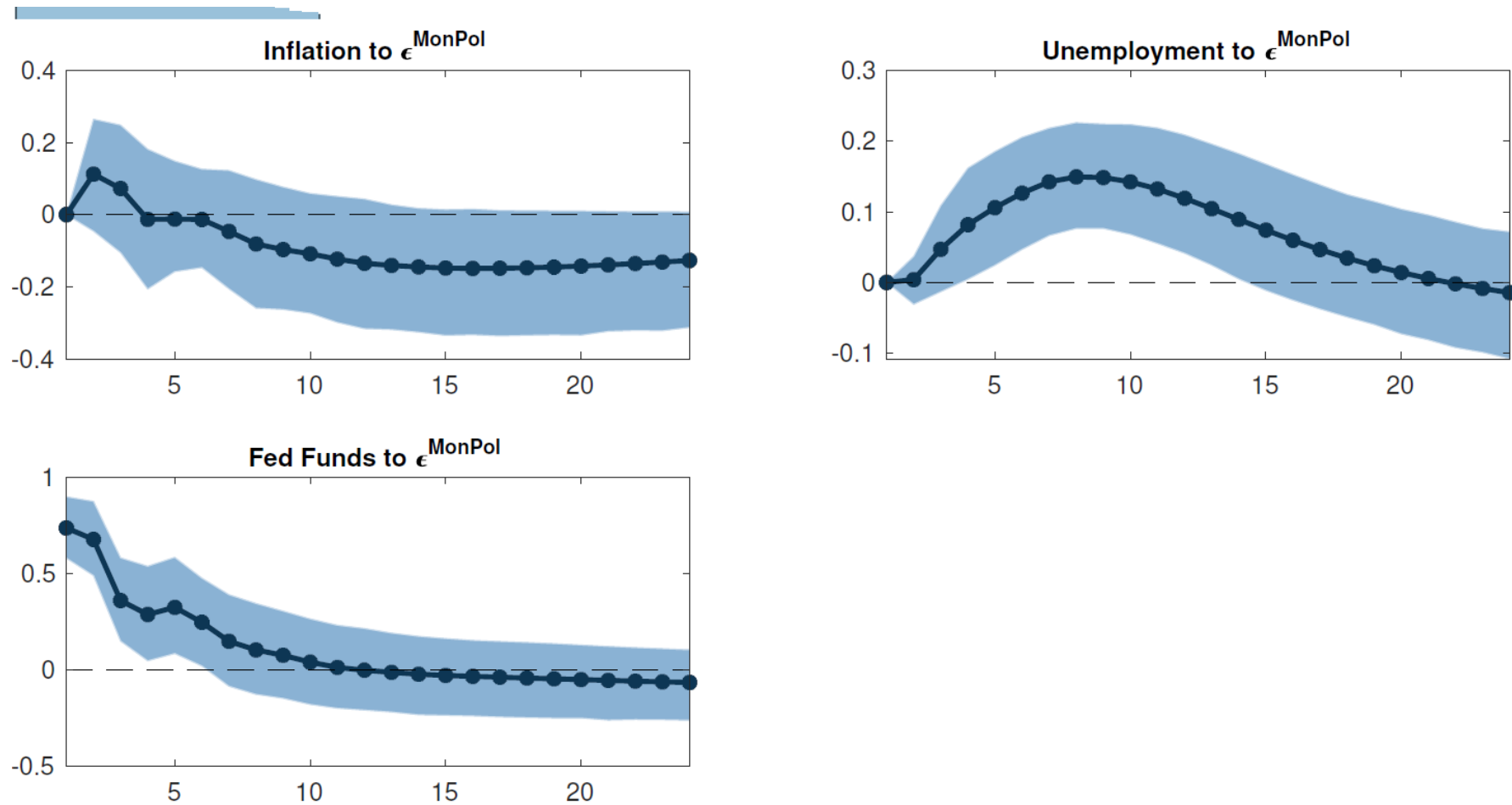


Figure: Impulse-response function

* Note *

Normalization in a reg. eqn.

$$\alpha^* y_t = \beta^* x_t + e_t \quad e_t \sim N(0, \sigma^{*2})$$

① 양변을 α^* 에 나누자.

$$\Rightarrow y_t = \frac{\beta^*}{\alpha^*} x_t + \frac{1}{\alpha^*} e_t$$

$$\frac{e_t}{\alpha^*} \sim N(0, \sigma^2)$$

$$\Rightarrow y_t = \beta x_t + e_t$$

$$\text{단 } \sigma^2 = \frac{\sigma^{*2}}{\alpha^{*2}}$$

① if

$$\sigma^{*2} = 1$$

$$\alpha^* = 1$$

$$\beta = 1$$

$$\beta^* = 1$$

$$\sigma^2 = 1$$

② if

$$\sigma^{*2} = 4$$

$$\alpha^* = 2$$

$$\beta = 1$$

$$\sigma^2 = 1$$

$$\beta^* = 2$$

!!!

x_t, y_t Structural model $\alpha^*, \beta^*, \sigma^{*2}$

- Reduced form (*) : $\hat{\beta}, \hat{\alpha}^2$ (2H) (3H)

how) Normalize !!

$$\text{① } \alpha^* = 1 \text{ } \Rightarrow \text{Inter!}$$

(unknown 2H)

$$\text{② } \sigma^{*2} = 1$$

" !

(" ")

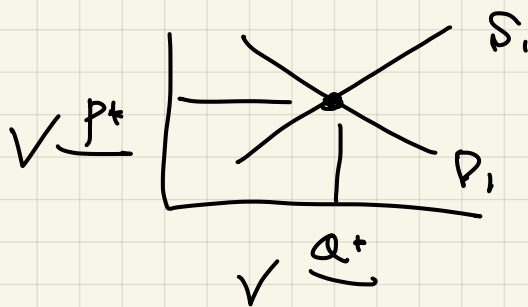
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연립 방정식 모형

$\Leftrightarrow$ VAR (Reduced - form)

(Dynamic Structural equation)

y_1 y_2
GDP $\frac{P}{Q}$
 y_{1t} y_{2t}



$$(D) \beta_{11} y_{1t} = -\beta_{12} y_{2t} + \alpha_{11} y_{1,t-1} + \alpha_{12} y_{2,t-1} + \varepsilon_{1t}, \quad \varepsilon_{1t} \sim \text{iid} (0,1)$$

$$(S) \beta_{21} y_{2t} = -\beta_{21} y_{1t} + \alpha_{21} y_{1,t-1} + \alpha_{22} y_{2,t-1} + \varepsilon_{2t}, \quad \varepsilon_{2t} \sim \text{iid} (0,1)$$

$$\Rightarrow \begin{bmatrix} \beta_{11} & \beta_{12} \\ \beta_{21} & \beta_{22} \end{bmatrix} \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \alpha_{11} & \alpha_{12} \\ \alpha_{21} & \alpha_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix}$$

$$\text{배치} \quad \beta \cdot Y_t = \alpha \cdot Y_{t-1} + \varepsilon_t + \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

양변에 β^{-1} 곱하자!

(I)

$\Rightarrow$

$$Y_t = \underbrace{\beta^{-1} \alpha}_{\text{I}} Y_{t-1} + \underbrace{\beta^{-1} \varepsilon_t}_{\varepsilon_t}$$

VAR //

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Structural VAR Model

Consider a (2×1) vector of zero-mean time series Y_t , composed of t observations and an initial condition Y_0

$$Y_t = \begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix}, \quad Y_0 = \begin{bmatrix} y_{10} \\ y_{20} \end{bmatrix}$$

Assume that the two time series in y_t are covariance stationary, which means (for $i = 1, 2$)

- ▶ Constant mean $\mathbb{E}[y_{it}] = \mu_i$
- ▶ Constant variance $\text{Var}[y_{it}] = \sigma_i^2$
- ▶ Constant autocovariance $\text{COV}[y_{it}, y_{it-j}] = \gamma_i(j)$

A structural VAR of order 1 is given by

$$Y_t = \Phi Y_{t-1} + B \varepsilon_t$$

where Φ and B are (2×2) matrices of model parameters; ε_t is an (2×1) vector of unobservable zero-mean white noise processes (shocks).

Structural VAR Model

The VAR(1) model $Y_t = \Phi Y_{t-1} + B\varepsilon_t$ is given by

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{1,t-1} \\ y_{2,t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

Or it can be written as a system of linear equations as

$$\begin{cases} y_{1t} = \phi_{11}y_{1,t-1} + \phi_{12}y_{2,t-1} + b_{11}\varepsilon_{1t} + b_{12}\varepsilon_{2t} \\ y_{2t} = \phi_{21}y_{1,t-1} + \phi_{22}y_{2,t-1} + b_{21}\varepsilon_{1t} + b_{22}\varepsilon_{2t} \end{cases}$$

Structural VAR Model

- ▶ The vector ε_t is a vector of unobservable zero mean white noise processes. They are something we cannot predict. So they are interpreted as shocks.
- ▶ The elements of ε_t are serially uncorrelated and independent of each other

That is,

$$\varepsilon_t = (\varepsilon'_{1t}, \varepsilon'_{2t})' \sim \mathcal{N}(0, I_2)$$

where

$$\text{Var}(\varepsilon_t) = \Sigma_\varepsilon = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad \text{and} \quad \text{CORR}(\varepsilon_t) = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

Structural VAR Model

Go back to our bivariate structural VAR(1)

$$\begin{bmatrix} y_{1t} \\ y_{2t} \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} y_{1t-1} \\ y_{2t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_{1t} \\ \varepsilon_{2t} \end{bmatrix}$$

- ▶ The structural VAR can be thought of as a description of the true structure of the economy (E.g.: an approximation of the structure imposed by a DSGE model)
- ▶ The structural shocks are shocks with a well-defined economic interpretation (E.g.: technology shocks or monetary policy shocks)
- ▶ As $\varepsilon_t \sim \mathcal{N}(0, I_2)$ we can move one shock keeping the other shocks fixed (That is: we can focus on the causal effect of one shock at the time)

Structural VAR Model

Suppose that y_{1t} and y_{2t} are output growth (z_t) and the policy rate (r_t), both demeaned.

- ▶ ε_{1t} and ε_{2t} are a demand shock ($\varepsilon_t^{\text{Demand}}$) and a monetary policy shock ($\varepsilon_t^{\text{MonPol}}$)
- ▶ Assume B is known.

Given the dynamics of the two variables,

$$\begin{bmatrix} z_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{\text{Demand}} \\ \varepsilon_t^{\text{MonPol}} \end{bmatrix}$$

- ▶ The coefficient b_{12} captures the (contemporaneous) '**marginal effect**' of a monetary policy shock on output growth
- ▶ The Φ matrix allows us to trace the 'dynamic marginal effect' of the monetary policy shock over time
- ▶ We can add more variables and look at other shocks: aggregate supply, oil price,...

Structural VAR Model

However, we cannot observe the structural shocks ε_t . In this case, how can we estimate B? What we can do is to 'bundle' the ε_t into a single object:

$$e_t = B\varepsilon_t \Rightarrow \begin{bmatrix} e_{zt} \\ e_{rt} \end{bmatrix}$$

$$\begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{\text{Demand}} \\ \varepsilon_t^{\text{MonPol}} \end{bmatrix} \Rightarrow \begin{cases} e_{zt} = b_{11}\varepsilon_t^{\text{Demand}} + b_{12}\varepsilon_t^{\text{MonPol}} \\ e_{rt} = b_{21}\varepsilon_t^{\text{Demand}} + b_{22}\varepsilon_t^{\text{MonPol}} \end{cases}$$

The VAR becomes

$$Y_t = \Phi Y_{t-1} + e_t$$

This model is called the 'reduced-form' VAR model. We can easily estimate Φ and e_t with OLS (where e_t will be OLS residuals).

Reduced-form VAR Model

The reduced-form VAR(1) is given by

$$Y_t = \Phi Y_{t-1} + e_t$$

In matrix form

$$\begin{bmatrix} z_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} e_{zt} \\ e_{rt} \end{bmatrix}$$

Or as a system of linear equations

$$\begin{cases} z_t = \phi_{11}z_{t-1} + \phi_{12}r_{t-1} + e_{zt} \\ r_t = \phi_{21}z_{t-1} + \phi_{22}r_{t-1} + e_{rt} \end{cases}$$

Reduced-form VAR Model

A key object of interest in the reduced-form VAR is the covariance matrix of the reduced-form residuals

$$\Sigma_e = \begin{bmatrix} \sigma_z^2 & \sigma_{zr} \\ \sigma_{zr} & \sigma_r^2 \end{bmatrix}$$

- ▶ Differently from the structural shocks (which are orthogonal), the reduced-form residuals are correlated among each other
- ▶ This is because the elements of e_t inherit all the contemporaneous relations among the endogenous variables Y_t

To see that, remember how the reduced form residuals are defined

$$\begin{cases} e_{zt} = b_{11}\varepsilon_t^{\text{Demand}} + b_{12}\varepsilon_t^{\text{MonPol}} \\ e_{rt} = b_{21}\varepsilon_t^{\text{Demand}} + b_{22}\varepsilon_t^{\text{MonPol}} \end{cases}$$

Reduced-form VAR Model

- ▶ To make causal inference (e.g. the effects on z_t of a shock to $\varepsilon_t^{\text{MonPol}}$) we need to find a way to recover B .
- ▶ We cannot perform causal inference with the reduced-form VAR because the reduced-form residuals are correlated among each other .

$$\Sigma_e = \begin{bmatrix} \sigma_z^2 & \sigma_{zr} \\ \sigma_{zr} & \sigma_r^2 \end{bmatrix}$$

- ▶ But we can estimate Σ_e using OLS residuals.
- ▶ Recovering B matrix from the estimated Σ_e is the essence of identification in VARs.

Identification

- ▶ We can estimate the reduced-form VAR with OLS. But we cannot estimate the structural VAR because of the unobserved structural shocks.
- ▶ Assume we already have an OLS estimate of $\hat{\Phi}$ and $\hat{e}_t$:

$$\begin{bmatrix} z_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} e_{zt} \\ e_{rt} \end{bmatrix}$$

- ▶ Question: What is the effect of a monetary policy shock on GDP growth? Unfortunately, the reduced form innovations (e_{zt} or e_{rt}) are not going to help us in answering the question

Identification

- To see that, assume that the 'true' (and unobserved) model of the economy is given by

$$\begin{bmatrix} z_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{\text{Demand}} \\ \varepsilon_t^{\text{MonPol}} \end{bmatrix}$$

- It is obvious that the reduced form innovations are a linear combination of the two structural shocks

$$e_{zt} = b_{11}\varepsilon_t^{\text{Demand}} + b_{12}\varepsilon_t^{\text{MonPol}}$$

$$e_{rt} = b_{21}\varepsilon_t^{\text{Demand}} + b_{22}\varepsilon_t^{\text{MonPol}}$$

- An increase in e_{rt} is not a monetary policy shock!

Identification

- It is obvious that the reduced form innovations are a linear combination of the two structural shocks

$$e_{zt} = b_{11}\varepsilon_t^{\text{Demand}} + b_{12}\varepsilon_t^{\text{MonPol}}$$

$$e_{rt} = b_{21}\varepsilon_t^{\text{Demand}} + b_{22}\varepsilon_t^{\text{MonPol}}$$

- An increase in e_{rt} could be due to
 - 1 A positive demand shock that increases both output growth and the policy rate ($b_{21} > 0$)
 - 2 Or a monetary policy shock that decreases output growth and increases the policy rate ($b_{22} > 0$)
- How to know whether is [1] or [2]? This is the very nature of the identification problem!

Identification

- ▶ The identification problem consists in finding a mapping from the reduced form VAR to its structural counterpart

$$e_t = B\varepsilon_t$$

- ▶ To do that, we can exploit the relation between reduced form and structural innovations to write

$$\Sigma_e = \mathbb{E} [e_t e_t'] = \mathbb{E} [B\varepsilon_t (B\varepsilon_t)'] = B\mathbb{E} (\varepsilon_t \varepsilon_t') B' = B\Sigma_\varepsilon B' = BB'$$

- ▶ The identification problem simply boils down to finding a B matrix that satisfies $\Sigma_e = BB'$
- ▶ Unfortunately this is not as easy as it sounds. Why?
- ▶ Hint: There are infinite combinations of B that give you the same Σ_e

Identification

- Think of $\Sigma_e = BB'$ as a system of equations

$$\begin{bmatrix} \sigma_z^2 & \sigma_{zr} \\ - & \sigma_r^2 \end{bmatrix} = \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{21} \\ b_{12} & b_{22} \end{bmatrix}$$

- Can be rewritten as

$$\begin{cases} \sigma_z^2 = b_{11}^2 + b_{12}^2 \\ \sigma_{zr} = b_{11}b_{21} + b_{12}b_{22} \\ \sigma_{zr} = b_{11}b_{21} + b_{12}b_{22} \\ \sigma_r^2 = b_{21}^2 + b_{22}^2 \end{cases}$$

- Problem: Because of the symmetry of the Σ_e matrix, the second and the third equation are identical
- We are left with 4 unknowns (the elements of B) but only 3 equations!

Identification: Summary

- ▶ Identification $\rightarrow$ Find a B that satisfies $\Sigma_e = BB'$
- ▶ There are infinite of such B 's
- ▶ In our simple example, we have to solve a system of 3 equations in 4 unknowns. How can we do it? Add one more equation. Or equivalently, restrict one structural parameter to a specific value.
- ▶ Economic theory can help in providing the 'missing' equation
- ▶ Make an assumption about the structure of the economy based on your beliefs (e.g. long-run monetary neutrality)
- ▶ Try to map this assumption into an equation that involves the VAR parameters
- ▶ The additional equation is known as a restriction
- ▶ That is: the additional equation restricts the set of infinite B matrices to a single one (or few ones) that are consistent with your assumption

Identification Strategy

- ▶ Identification is achieved by assuming that some shocks have zero contemporaneous effect on some of the endogenous variables
- ▶ Reference 1: Sims, Christopher A. "Macroeconomics and reality." *Econometrica*(1980)
- ▶ Reference 2: Christiano, Lawrence J., Martin Eichenbaum, and Charles L. Evans. "Monetary policy shocks: What have we learned and to what end?." *Handbook of macroeconomics* 1 (1999)
- ▶ For example, assume that monetary policy works with a lag and has no contemporaneous effects on output
- ▶ But how can we impose restrictions on the effect of a structural shock?

Identification Strategy

- ▶ Solution: Impose zero restrictions on the impact matrix B
- ▶ The b_{12} coefficient captures the contemporaneous effect of monetary policy on output growth

$$\begin{bmatrix} z_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} b_{11} & 0 \\ b_{21} & b_{22} \end{bmatrix} \begin{bmatrix} \varepsilon_t^{\text{Demand}} \\ \varepsilon_t^{\text{Monpol}} \end{bmatrix}$$

- ▶ Implication: We now have 3 structural parameters to estimate (instead of 4) and 3 restrictions implied by Σ_e
- ▶ This restriction allows us to solve the identification problem.
- ▶ In graduate school, we will be able to learn more identification strategies for structural VARs.

Identification Strategy

- ▶ Generally speaking, for exact identification, we need total $n(n - 1)/2$ restrictions to identify all structural shocks in a VAR system with n variables.
- ▶ e.g.

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Structural VAR Model

- ▶ Let $\mathbf{Y}_t = (y_{1t}, y_{2t}, \dots, y_{nt})'$ denote an $(n \times 1)$ vector of time series variables.
- ▶ The basic p -lag structural vector autoregressive (VAR(p)) model has the form

$$\mathbf{Y}_t = \mathbf{c} + \Phi_1 \mathbf{Y}_{t-1} + \Phi_2 \mathbf{Y}_{t-2} + \dots + \Phi_p \mathbf{Y}_{t-p} + \mathbf{B} \varepsilon_t,$$
$$\varepsilon_t \sim \text{i.i.d.}(\mathbf{0}, \mathbf{I}_n)$$

- ▶ All structural shocks are assumed to be independent from each other.
- ▶ Note that serial correlation should not exist in the structural shocks. $\text{Cov}(\varepsilon_t, \varepsilon_{t-j}) = 0$ for $j = 1, 2, \dots$. This is not an assumption. We can statistically check it using corresponding reduced-form shocks.

Reduced-form VAR Model

- ▶ The basic p -lag reduced-form vector autoregressive (VAR(p)) model has the form

$$\mathbf{Y}_t = \mathbf{c} + \Phi_1 \mathbf{Y}_{t-1} + \Phi_2 \mathbf{Y}_{t-2} + \cdots + \Phi_p \mathbf{Y}_{t-p} + e_t,$$
$$e_t \sim \text{i.i.d.}(\mathbf{0}, \Sigma)$$

- ▶ Note that serial correlation should not in the reduced-form error vector. $\text{Cov}(e_t, e_{t-j}) = 0$ for $j = 1, 2, \dots$
- ▶ The relation between the reduced-form and structural VARs are defined by $e_t = \mathbf{B}\varepsilon_t$.
- ▶ And this relation indicates that $\text{Var}[e_t] = \Sigma = \text{Var}[\mathbf{B}\varepsilon_t] = \mathbf{B}\mathbf{I}_n\mathbf{B}' = \mathbf{B}\mathbf{B}'$.

Model Extension

The basic VAR(p) model may be too restrictive to sufficiently represent the main characteristics of the data. The general form of the VAR(p) model with deterministic terms and strictly exogenous variables $\mathbf{X}_t$ is given by

$$\mathbf{Y}_t = \Phi_1 \mathbf{Y}_{t-1} + \Phi_2 \mathbf{Y}_{t-2} + \cdots + \Phi_p \mathbf{Y}_{t-p} + \boldsymbol{\Phi} \mathbf{D}_t + \mathbf{G} \mathbf{X}_t + e_t$$

$\mathbf{D}_t$ = deterministic terms like an intercept term and a linear time trend

$\mathbf{X}_t$ = contemporaneously exogenous variables ($E[e_t | \mathbf{X}_t] = \mathbf{0}$)

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Estimation

Recall that our simple example is the following bi-variate (reduced-form) VAR(1):

$$Y_t = \Phi Y_{t-1} + e_t$$

In matrix form

$$\begin{bmatrix} z_t \\ r_t \end{bmatrix} = \begin{bmatrix} \phi_{11} & \phi_{12} \\ \phi_{21} & \phi_{22} \end{bmatrix} \begin{bmatrix} z_{t-1} \\ r_{t-1} \end{bmatrix} + \begin{bmatrix} e_{zt} \\ e_{rt} \end{bmatrix}$$

Or as a system of linear equations

$$\begin{cases} z_t = \phi_{11}z_{t-1} + \phi_{12}r_{t-1} + e_{zt} \\ r_t = \phi_{21}z_{t-1} + \phi_{22}r_{t-1} + e_{rt} \end{cases}$$

Estimation

The model is composed of two linear regression models although two reduced-form shocks are correlated:

$$\begin{cases} z_t = \phi_{11}z_{t-1} + \phi_{12}r_{t-1} + e_{zt} \\ r_t = \phi_{21}z_{t-1} + \phi_{22}r_{t-1} + e_{rt} \end{cases}$$

- ▶ No problem using OLS for estimation.
- ▶ Estimate the first equation using OLS. Obtain residuals, $\hat{e}_{zt}$
- ▶ Estimate the second equation using OLS. Obtain residuals, $\hat{e}_{rt}$
- ▶ Estimate the covariance matrix $\Sigma = \text{Var}[E_t]$ where $E_t = [e_{zt}, e_{rt}]'$ using the following sample covariance matrix estimator:

$$\hat{\Sigma}_{(2 \times 2)} = \frac{1}{T - k} \sum_{t=1}^T \hat{\mathbf{E}}_t \hat{\mathbf{E}}_t', \quad \hat{\mathbf{E}}_t = [\hat{e}_{zt}, \hat{e}_{rt}]'$$

- ▶ k is the total number of coefficients in VAR. Here, we have $k = 4$.

Estimation

Assume that the $\text{VAR}(p)$ model is covariance stationary, and there are no restrictions on the parameters of the model. Each equation in the $\text{VAR}(p)$ may be written as

$$\mathbf{y}_i = \mathbf{Z}\boldsymbol{\beta}_i + \mathbf{e}_i, i = 1, \dots, n$$

- ▶ $\mathbf{y}_i$ is a $(T \times 1)$ vector of observations on the i^{th} equation
- ▶ $\mathbf{Z}$ is a $(T \times k)$ matrix with t^{th} row given by $\mathbf{Z}'_t = (1, \mathbf{Y}'_{t-1}, \dots, \mathbf{Y}'_{t-p})$ and $k = np + 1$
- ▶ $\boldsymbol{\beta}_i$ is a $(k \times 1)$ vector of VAR parameters for the i th equation and $\mathbf{e}_i$ is a $(T \times 1)$ error with covariance matrix $\sigma_i^2 \mathbf{I}_T$.

Estimation

Assume that the $\text{VAR}(p)$ model is covariance stationary, and there are no restrictions on the parameters of the model. Each equation in the $\text{VAR}(p)$ may be written as

$$\mathbf{y}_i = \mathbf{Z}\boldsymbol{\beta}_i + \mathbf{e}_i, i = 1, \dots, n$$

- ▶ Under the classical model assumptions, OLS estimator $\hat{\boldsymbol{\beta}}_i = (\mathbf{Z}'\mathbf{Z})^{-1} (\mathbf{Z}'\mathbf{y}_i)$ is BLUE. (Best Linear Unbiased Estimator)
- ▶ The residual vector is given by $\hat{\mathbf{E}}_t^{(n \times 1)} = \mathbf{Y}_t - \hat{\mathbf{B}}'\mathbf{Z}_t = [\hat{\mathbf{e}}_1, \dots, \hat{\mathbf{e}}_n]'$. Then the covariance matrix is estimated as

$$\hat{\boldsymbol{\Sigma}}_{(n \times n)} = \frac{1}{T - k} \sum_{t=1}^T \hat{\mathbf{E}}_t \hat{\mathbf{E}}_t'$$

- ▶ MLE can be used for estimation as well. But MLE could be numerically unstable because there are many model parameters to estimate.

Lag Length Selection

The lag length for the $\text{VAR}(p)$ model may be determined using model selection criteria. The general approach is to fit $\text{VAR}(p)$ models with orders $p = 0, \dots, p_{\max}$ and choose the value of p which minimizes some model selection criteria

$$\text{MSC}(p) = \ln |\tilde{\Sigma}(p)| + c_T \cdot \varphi(n, p)$$

$$\tilde{\Sigma}(p) = T^{-1} \sum_{t=1}^T \hat{\mathbf{E}}_t \hat{\mathbf{E}}_t'$$

c_T = function of sample size

$\varphi(n, p)$ = penalty function

Information Criteria

The three most common information criteria are the Akaike (AIC), SchwarzBayesian (BIC):

$$\text{AIC}(p) = \ln |\tilde{\Sigma}(p)| + \frac{2}{T}pn^2$$

$$\text{BIC}(p) = \ln |\tilde{\Sigma}(p)| + \frac{\ln T}{T}pn^2$$

- ▶ AIC criterion asymptotically overestimates the order with positive probability.
- ▶ BIC criteria estimate the order consistently under fairly general conditions if the true order p is less than or equal to $p_{\max}$.

Outline

Motivation

A Simple Example

Model Specification

Estimation

Forecasting with Reduced-form VAR

Impulse-Response analysis with Structural VAR

Granger Causality

Implementation in Python

Forecasting

- The first order dynamic equation in matrix form (it is called 'state-space representation') for the reduced-form VAR(p) model is given by

$$\begin{pmatrix} \mathbf{Y}_t \\ \mathbf{Y}_{t-1} \\ \vdots \\ \mathbf{Y}_{t-p+1} \end{pmatrix} = \begin{pmatrix} \mathbf{c} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix} + \begin{pmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_p \\ \mathbf{I}_n & \mathbf{0} & \vdots & \mathbf{0} \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \cdots & \mathbf{I}_n & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{Y}_{t-1} \\ \mathbf{Y}_{t-2} \\ \vdots \\ \mathbf{Y}_{t-p} \end{pmatrix} + \begin{pmatrix} e_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix}$$
$$\boldsymbol{\zeta}_t = \mathbf{C} + \mathbf{F}\boldsymbol{\zeta}_{t-1} + \mathbf{v}_t$$

Forecasting

- The first order dynamic equation in matrix form (it is called 'state-space representation') for the reduced-form VAR(p) model is given by

$$\boldsymbol{\zeta}_t = \mathbf{C} + \mathbf{F}\boldsymbol{\zeta}_{t-1} + \mathbf{v}_t$$
$$Var[\mathbf{v}_t] = \begin{pmatrix} \Sigma & \mathbf{0} & \dots & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \vdots & \mathbf{0} \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \dots & \mathbf{0} & \mathbf{0} \end{pmatrix}$$

Forecasting

- ▶ Let's suppose that we have the estimated model parameters of the reduced-form VAR. Given the estimated model parameters, how do we forecast?
- ▶ Forecasting from a reduced-form VAR(p) is a straightforward extension of forecasting from an AR(p).
- ▶ Using the state-space framework, we can rewrite a reduced-form VAR(p) model as

$$\boldsymbol{\zeta}_t = \mathbf{C} + \mathbf{F}\boldsymbol{\zeta}_{t-1} + \mathbf{v}_t.$$

- ▶ Thus,

$$\begin{aligned}\boldsymbol{\zeta}_{t+j} &= \mathbf{C} + \mathbf{F}\boldsymbol{\zeta}_{t+j-1} + \mathbf{v}_{t+j} \\ &= (\mathbf{I} + \mathbf{F})\mathbf{C} + \mathbf{F}^2\boldsymbol{\zeta}_{t+j-2} + \mathbf{v}_{t+j} + \mathbf{F}\mathbf{v}_{t+j-1} \\ &\vdots \\ &= (\mathbf{I} + \mathbf{F} + \dots + \mathbf{F}^{j-1})\mathbf{C} + \mathbf{F}^j\boldsymbol{\zeta}_t \\ &\quad + \mathbf{v}_{t+j} + \mathbf{F}\mathbf{v}_{t+j-1} + \dots + \mathbf{F}^{j-1}\mathbf{v}_{t+1}\end{aligned}$$

Forecasting

- ▶ $E[\boldsymbol{\zeta}_{t+j}|\mathcal{I}_t] = (\mathbf{I} + \mathbf{F} + \dots \mathbf{F}^{j-1}) \mathbf{C} + \mathbf{F}^j \boldsymbol{\zeta}_t$ where $\mathcal{I}_t$ is information set up to time t .
- ▶ $Var[\boldsymbol{\zeta}_{t+j}|\mathcal{I}_t] = Var[\mathbf{v}_{t+j}] + \mathbf{F} Var[\mathbf{v}_{t+j}] \mathbf{F}' + \dots + \mathbf{F}^{j-1} Var[\mathbf{v}_{t+1}] (\mathbf{F}^{j-1})'$ where $\mathcal{I}_t$ is information set up to time t .

Outline

Motivation

A Simple Example

Model Specification

Estimation

Forecasting with Reduced-form VAR

Impulse-Response analysis with Structural VAR

Granger Causality

Implementation in Python

Reduced-form VAR Model

- ▶ The basic p -lag reduced-form vector autoregressive (VAR(p)) model has the form

$$\mathbf{Y}_t = \mathbf{c} + \Phi_1 \mathbf{Y}_{t-1} + \Phi_2 \mathbf{Y}_{t-2} + \cdots + \Phi_p \mathbf{Y}_{t-p} + e_t,$$
$$e_t \sim \text{i.i.d.}(\mathbf{0}, \mathbf{\Sigma})$$

- ▶ Note that serial correlation should not exist in the reduced-form error vector. $\text{Cov}(e_t, e_{t-j}) = 0$ for $j = 1, 2, \dots$
- ▶ The relation between the reduced-form and structural VARs are defined by $e_t = \mathbf{B}\varepsilon_t$.
- ▶ And this relation indicates that $\text{Var}[e_t] = \mathbf{\Sigma} = \text{Var}[\mathbf{B}\varepsilon_t] = \mathbf{B}\mathbf{I}_n\mathbf{B}' = \mathbf{B}\mathbf{B}'$.

Wold Representation Theorem

- ▶ Let's suppose that we have the estimated model parameters of the reduced-form VAR. Given the estimated model parameters, how do we do causal inference?
- ▶ The relation between the reduced-form and structural VARs are defined by $e_t = \mathbf{B}\varepsilon_t$.
- ▶ By the Wold representation theorem, we know that there exists the following Wold form for the reduced-form VAR model

$$\mathbf{Y}_t = \boldsymbol{\mu} + e_t + \boldsymbol{\Psi}_1 e_{t-1} + \cdots + \boldsymbol{\Psi}_j e_{t-j} + \cdots$$

where $\boldsymbol{\mu} = E[\mathbf{Y}_t]$ and e_t is the prediction errors (reduced-form VAR shocks).

- ▶ There are infinitely many Wold coefficients $\boldsymbol{\Psi}_j$. So we approximate the Wold representation with a finite order reduced-form VAR model.

Wold Representation Theorem

- ▶ Performing causal inference in terms of e_t does not make a sense because the reduce-form shocks are correlated.
- ▶ We need to perform causal inference in terms of ε_t because structural shocks are independent from each other.
- ▶ Thus, we rewrite the Wold representation as

$$\begin{aligned}\mathbf{Y}_t &= \boldsymbol{\mu} + e_t + \boldsymbol{\Psi}_1 e_{t-1} + \cdots + \boldsymbol{\Psi}_j e_{t-j} + \cdots \\ &= \boldsymbol{\mu} + \mathbf{B}\varepsilon_t + \boldsymbol{\Psi}_1 \mathbf{B}\varepsilon_{t-1} + \cdots + \boldsymbol{\Psi}_j \mathbf{B}\varepsilon_{t-j} + \cdots \\ &= \boldsymbol{\mu} + \boldsymbol{\Psi}_0^* \varepsilon_t + \boldsymbol{\Psi}_1^* \varepsilon_{t-1} + \cdots + \boldsymbol{\Psi}_j^* \varepsilon_{t-j} + \cdots\end{aligned}$$

where $\boldsymbol{\Psi}_k^* = \boldsymbol{\Psi}_k \mathbf{B}$ for $k = 0, 1, 2, \dots$ and $\boldsymbol{\Psi}_0 = \mathbf{I}_n$.

Impulse Response Analysis

$$\mathbf{Y}_t = \boldsymbol{\mu} + \boldsymbol{\Psi}_0^* \varepsilon_t + \boldsymbol{\Psi}_1^* \varepsilon_{t-1} + \cdots + \boldsymbol{\Psi}_j^* \varepsilon_{t-j} + \cdots$$

where

$$\boldsymbol{\Psi}_j^* \varepsilon_{t-j} = \begin{pmatrix} \psi_{11,j}^* & \psi_{12,j}^* & \cdots & \psi_{1n,j}^* \\ \psi_{21,j}^* & \psi_{22,j}^* & \cdots & \psi_{2n,j}^* \\ \vdots & \vdots & \ddots & \vdots \\ \psi_{n1,j}^* & \psi_{n2,j}^* & \cdots & \psi_{nn,j}^* \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t-j} \\ \varepsilon_{2,t-j} \\ \vdots \\ \varepsilon_{n,t-j} \end{pmatrix}$$

- From the Wold representation, we know the impulse response coefficients or marginal effects of a structural shocks are given by

$$\frac{\partial y_{i,t}}{\partial \varepsilon_{h,t-j}} = \frac{\partial y_{i,t+j}}{\partial \varepsilon_{h,t}} = \psi_{ih,j}^*$$

for $i, h = 1, 2, \dots, n$.

Wold Representation Theorem

- ▶ Thus, we rewrite the Wold representation as

$$\begin{aligned}\mathbf{Y}_t &= \boldsymbol{\mu} + e_t + \boldsymbol{\Psi}_1 e_{t-1} + \cdots + \boldsymbol{\Psi}_j e_{t-j} + \cdots \\ &= \boldsymbol{\mu} + \mathbf{B}\varepsilon_t + \boldsymbol{\Psi}_1 \mathbf{B}\varepsilon_{t-1} + \cdots + \boldsymbol{\Psi}_j \mathbf{B}\varepsilon_{t-j} + \cdots \\ &= \boldsymbol{\mu} + \boldsymbol{\Psi}_0^* \varepsilon_t + \boldsymbol{\Psi}_1^* \varepsilon_{t-1} + \cdots + \boldsymbol{\Psi}_j^* \varepsilon_{t-j} + \cdots\end{aligned}$$

where $\boldsymbol{\Psi}_k^* = \boldsymbol{\Psi}_k \mathbf{B}$ for $k = 0, 1, 2, \dots$ and $\boldsymbol{\Psi}_0 = \mathbf{I}_n$.

- ▶ For causal inference (impulse response analysis), we need two things, $\mathbf{B}$ and $\boldsymbol{\Psi}_j$.

Recovering Wold Coefficients Ψ_j

- From the reduced-form VAR model, we can obtain the state space representation

$$\begin{pmatrix} \mathbf{Y}_t \\ \mathbf{Y}_{t-1} \\ \vdots \\ \mathbf{Y}_{t-p+1} \end{pmatrix} = \begin{pmatrix} \mathbf{c} \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix} + \begin{pmatrix} \Phi_1 & \Phi_2 & \cdots & \Phi_p \\ \mathbf{I}_n & \mathbf{0} & \vdots & \mathbf{0} \\ \vdots & \ddots & \vdots & \vdots \\ \mathbf{0} & \cdots & \mathbf{I}_n & \mathbf{0} \end{pmatrix} \begin{pmatrix} \mathbf{Y}_{t-1} \\ \mathbf{Y}_{t-2} \\ \vdots \\ \mathbf{Y}_{t-p} \end{pmatrix} + \begin{pmatrix} e_t \\ \mathbf{0} \\ \vdots \\ \mathbf{0} \end{pmatrix}$$
$$\boldsymbol{\zeta}_t = \mathbf{C} + \mathbf{F}\boldsymbol{\zeta}_{t-1} + \mathbf{v}_t$$

- By repeated substitutions, we can obtain

$$\boldsymbol{\zeta}_t = \dots \mathbf{C} + \mathbf{v}_t + \mathbf{F}\mathbf{v}_{t-1} + \dots \mathbf{F}^j \mathbf{v}_{t-j} + \dots$$

Recovering Wold Coefficients Ψ_j

- ▶ By repeated substitutions, we can obtain

$$\xi_t = \dots \mathbf{C} + \mathbf{v}_t + \mathbf{F}\mathbf{v}_{t-1} + \dots \mathbf{F}^j \mathbf{v}_{t-j} + \dots$$

- ▶ Because the first n elements of ξ_t is $\mathbf{Y}_t$ and the first n elements of $\mathbf{v}_t$ is e_t , Ψ_j is $\mathbf{F}_{[1:n,1:n]}^j$ which is the first n by n sub-matrix of $\mathbf{F}^j$.

Recovering **B**

- ▶ We need identification assumption. The most commonly used assumption is a recursive identification assumption: **B** is a lower triangular matrix.
- ▶ The first structural shock $\varepsilon_{1,t}$ affects all endogenous variables contemporaneously.
- ▶ The second structural shock $\varepsilon_{2,t}$ affects all endogenous variables contemporaneously except $y_{1,t}$.
- ▶ The third structural shock $\varepsilon_{3,t}$ affects all endogenous variables contemporaneously except $y_{1,t}$ and $y_{2,t}$.
- ▶ ...
- ▶ The last structural shock $\varepsilon_{n,t}$ only affects $y_{n,t}$ contemporaneously.

Recovering **B**

- ▶ Suppose that the recursive identification assumption is imposed over a tri-variate VAR(p) model.

$$\begin{pmatrix} UMR_t \\ IFR_t \\ FFR_t \end{pmatrix} = \dots + \begin{pmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \end{pmatrix}$$

where UMR_t is the unemployment rate, IFR_t is the inflation rate, FFR_t is the federal funds rate.

- ▶ $\varepsilon_{1,t}$ affects all variables at time t . $\varepsilon_{2,t}$ affects IFR_t and FFR_t at time t . $\varepsilon_{3,t}$ affects IFR_t at time t .
- ▶ $\varepsilon_{1,t}$ can be interpreted as a demand shock. $\varepsilon_{2,t}$ can be interpreted as a supply shock. $\varepsilon_{3,t}$ can be interpreted as a monetary policy shock.
- ▶ The third structural shock $\varepsilon_{3,t}$ affects all endogenous variables contemporaneously except $y_{1,t}$ and $y_{2,t}$.

Recovering **B**

$$\begin{pmatrix} UMR_t \\ IFR_t \\ FFR_t \end{pmatrix} = \dots + \begin{pmatrix} b_{11} & 0 & 0 \\ b_{21} & b_{22} & 0 \\ b_{31} & b_{32} & b_{33} \end{pmatrix} \begin{pmatrix} \varepsilon_{1,t} \\ \varepsilon_{2,t} \\ \varepsilon_{3,t} \end{pmatrix}$$

- ▶ Alternatively, we can say that UMR_t is the most exogenous variable in the VAR system and FFR_t is the most endogenous variable in the VAR system.
- ▶ Alternatively, we can say that UMR_t is the slowest variable in the VAR system and FFR_t is the fastest variable in the VAR system.

Recovering **B**

- ▶ Apply the Cholesky decomposition to Σ to obtain the **unique** lower triangular matrix **H**:

$$\Sigma = \mathbf{H}\mathbf{H}'$$

- ▶ The Cholesky decomposition is similar to square root operation.
- ▶ The structural VAR model implies that

$$\Sigma = \mathbf{B}\mathbf{B}'$$

- ▶ Step 1: Estimate Σ . Let $\hat{\Sigma}$ denote the estimated covariance matrix for the reduced-form VAR errors.
- ▶ Step 2: Apply the Cholesky decomposition to $\hat{\Sigma}$ and obtain the unique lower triangular matrix, $\hat{\mathbf{H}}$.
- ▶ Step 3: We are done. Our estimate for **B** is $\hat{\mathbf{H}}$.

Outline

Motivation

A Simple Example

Model Specification

Estimation

Forecasting with Reduced-form VAR

Impulse-Response analysis with Structural VAR

Granger Causality

Implementation in Python

Granger Causality

One of the main uses of VAR models is forecasting. The following intuitive notion of a variable's forecasting ability is due to Granger (1969).

- ▶ If a variable, or group of variables, y_1 is found to be helpful for predicting another variable, or group of variables, y_2 then y_1 is said to Granger-cause y_2 ; otherwise it is said to fail to Granger-cause y_2 .
- ▶ Formally, y_1 fails to Granger-cause y_2 if for all $s > 0$ the MSE of a forecast of $y_{2,t+s}$ based on $(y_{2,t}, y_{2,t-1}, \dots)$ is the same as the MSE of a forecast of $y_{2,t+s}$ based on $(y_{2,t}, y_{2,t-1}, \dots)$ and $(y_{1,t}, y_{1,t-1}, \dots)$.
- ▶ The notion of Granger causality does not imply true causality. It only implies forecasting ability.

Granger Causality

In a bivariate VAR(p), y_2 fails to Granger-cause y_1 if all of the p VAR coefficient matrices $\Phi_1, \dots, \Phi_p$ are lower triangular:

$$\begin{pmatrix} y_{1t} \\ y_{2t} \end{pmatrix} = \begin{pmatrix} c_1 \\ c_2 \end{pmatrix} + \begin{pmatrix} \phi_{11,1} & 0 \\ \phi_{21,1} & \phi_{22,1} \end{pmatrix} \begin{pmatrix} y_{1t-1} \\ y_{2t-1} \end{pmatrix} + \dots \\ + \begin{pmatrix} \phi_{11,p} & 0 \\ \phi_{21,p} & \phi_{22,p} \end{pmatrix} \begin{pmatrix} y_{1t-p} \\ y_{2t-p} \end{pmatrix} + \begin{pmatrix} e_{1t} \\ e_{2t} \end{pmatrix}$$

- ▶ Similarly, y_1 fails to Granger-cause y_2 if all of the coefficients on lagged values of y_1 are zero in the equation for y_2 .
- ▶ Notice that if y_2 fails to Granger-cause y_1 and y_1 fails to Granger-cause y_2 , then the VAR coefficient matrices $\Pi_1, \dots, \Pi_p$ are diagonal.
- ▶ We can test for Granger Causality.
 $H_0 : \phi_{12,1} = \phi_{12,2} = \dots = \phi_{12,p} = 0$, $H_a : H_0$ is not true.

Outline

Motivation

A Simple Example

Model Specification

Estimation

Forecasting with Reduced-form VAR

Impulse-Response analysis with Structural VAR

Granger Causality

Implementation in Python

VAR Estimation

The following Python code illustrates how to use a Python VAR library. The full code is provided in the class website.

```
1 # import necessary packages and libraries
2 import statsmodels.api as sm
3 from statsmodels.tsa.api import VAR
4 from statsmodels.tsa.vector_ar.var_model import
    VARResults
5
6
7 # If you would like to use a VAR model for data y (T \
    times n),
8 # Specify the VAR model
9 VARmodel = VAR(y)
10 # Select lag order
11 lag_order = VARmodel.select_order(maxlags=12)
12 # Report lag order result
13 lag_order.summary()
```

VAR Estimation

The following Python code illustrates how to use a Python VAR library. The full code is provided in the class website.

```
1 # Estimate the VAR model for the training set
2 results = VARmodel.fit(maxlags=12, ic='aic')
3 results.summary()
4
5 # Get the predicted values for the VAR model in the
   evaluation set
6 lag_order_aic = results.k_ar
7 lagged_values = train_data.values[-lag_order_aic:]
8
9 # Forecast for next 8 periods
10 forecasted_values = results.forecast(y=lagged_values,
   steps=8)
11 forecast_cov = results.forecast_cov(steps=8)
```

End.